

Problem J

Crystal Combination

Time limit: 1 second

Problem Description

Elaina, a magic teacher, has prepared an exam with t questions. In each question, given a target magic power x , a student must combine exactly two magic crystals with powers 2^a and 2^b for some non-negative integers a and b (possibly $a = b$), and report the minimum possible difference between the combined power and x , that is, $|2^a + 2^b - x|$. Please help the students finish the exam.

Input Format

Each test contains multiple test cases. The first line contains the number of test cases t . The description of the test cases follows.

The only line of each test case contains one integer x , representing the target magic power.

Output Format

For each test case, output one integer: the minimum possible difference between the combined power and x , that is, $|2^a + 2^b - x|$.

Technical Specification

- $1 \leq t \leq 10^4$
- $1 \leq x \leq 10^{18}$

Sample Input 1

```
3
7
10
1
```

Sample Output 1

```
1
0
1
```

$$|2^2 + 2^2 - 7| = 1$$

$$|2^1 + 2^3 - 10| = 0$$

$$|2^0 + 2^0 - 1| = 1$$

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